

w o/w data_storage_address	w o/w indices_address	w o/w world_address
1.40225E+14	1.40225E+14	1.40724E+14
1.40225E+14	1.40225E+14	1.40724E+14
1.40225E+14	1.40225E+14	1.40724E+14
1.40225E+14	1.40225E+14	1.40724E+14
1.40225E+14	1.40225E+14	1.40724E+14
1.40225E+14	1.40225E+14	1.40724E+14
1.40225E+14	1.40225E+14	1.40724E+14
1.40225E+14	1.40225E+14	1.40724E+14
1.40225E+14	1.40225E+14	1.40724E+14
Row1	L1 Cache	L2 Cache
Set Position data_storage	0	528
Set Position indices	0	464
Set Position world	62.125	46.125
Row2	L1 Cache	L2 Cache
Set Position data_storage	0	400
Set Position indices	0	336
Set Position world	62.125	46.125
Row3	L1 Cache	L2 Cache
Set Position data_storage	0	272
Set Position indices	0	208
Set Position world	62.125	46.125
Row4	L1 Cache	L2 Cache
Set Position data_storage	0	16
Set Position indices	0	952
Set Position world	62.125	46.125
Row5	L1 Cache	L2 Cache
Set Position data_storage	0	696
Set Position indices	0	632
Set Position world	62.125	46.125
Row6	L1 Cache	L2 Cache
Set Position data_storage	0	568
Set Position indices	0	504
Set Position world	62.125	46.125
Row7	L1 Cache	L2 Cache
Set Position data_storage	0	440
Set Position indices	0	376
Set Position world	62.125	46.125

Indices Address Delta	Indice+Data page size	Data Page Size bound	Indice Page
-8192	8192	TRUE	TRUE
-8192	8192	TRUE	TRUE
-16384	8192	TRUE	TRUE
-20480	16384	TRUE	TRUE
-8192	20480	TRUE	TRUE
-8192	8192	TRUE	TRUE
-8192	8192	TRUE	TRUE
-8192	8192	TRUE	TRUE
-36864	8192	TRUE	TRUE
	36864	TRUE	TRUE

L1 Cache	L2 Cache	L3 Cache
32768	512000	33554432
8	8	16
64	64	64
64	1000	32768
4096	64000	2097152

mmap or virtual alloc (4096 aligned) will cause eviction on the L1 cache. To utilize the full L1
er of ways of mmap/virtual alloc. To Avoid thrashing we can reuse mmap or virtual alloc
Each 64 byte offset of 4096 will map to the next set
So the hardware prefetches in set not ways

On AMD Zen processors, the L1D cache is
virtually indexed and physically tagged (VIPT).
we use the offset (use to point to the first byte in cache) +
index which is used to get the set in the cache
Both use the virtual address

we use the tag to get the specific way in a given set
uses the physical address

source: <https://gruss.cc/files/takeaway.pdf>

size bound

Row8	L1 Cache	L2 Cache
Set Position data_storage	0	312
Set Position indices	0	248
Set Position world	62.125	46.125
Row9	L1 Cache	L2 Cache
Set Position data_storage	0	736
Set Position indices	0	672
Set Position world	62.125	46.125

L3 Cache

26368
26304
4350.125

L3 Cache

25792
25728
4350.125



